

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 2: Image Processing Pipeline

QUESTION

What is the purpose of pre-processing?

ANSWER & EXPLANATION

The primary purpose of pre-processing in an image processing pipeline is to improve the quality of raw image data by suppressing unwanted distortions, enhancing specific visual features, and standardizing the data format so that downstream computer vision tasks (such as segmentation, feature extraction, object detection, and classification) can perform more accurately and efficiently.

KEY OBJECTIVES OF PRE-PROCESSING

- **Noise Reduction:** Removing random variations in intensity or color introduced during image capture (e.g., Gaussian noise, salt-and-pepper noise) using filtering techniques like median or Gaussian filters.
- **Contrast & Brightness Enhancement:** Adjusting dynamic range and luminance levels (e.g., via histogram equalization) to make low-visibility details legible.
- **Data Standardization & Normalization:** Resizing images to uniform dimensions, converting color spaces (e.g., RGB to Grayscale), and scaling pixel values to a standard range (e.g., $[0, 1]$) for neural network compatibility.
- **Geometric & Radiometric Correction:** Correcting optical lens distortions, rotation, perspective skew, and uneven illumination across the field of view.

CONCLUSION

Ultimately, pre-processing acts as a data-conditioning step that isolates signal from noise, ensuring that higher-level analysis algorithms focus on meaningful feature patterns rather than sensor artifacts.